

PAPER_1511_DPM_LAYER_WEIGHT_I6_DECOMPOSITION

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PAPER_1511 — DPM Layer-Weight $w_i = i^2 \times i \times i^3 = i^6$ EXACT Decomposition

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket D (Particle Physics) / Bucket A (Foundational)

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Status: CLOSED — Structural origin of i^6 amplification weight

Observation

PAPER_1155 (DPM 26-Layer Amplification) decomposes the layer- i weighting that produces $A_{26} = \sum i^6 = 1,307,797,101$ into a product of three physical fields:

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$$\begin{aligned} w_i &= [\text{SCm}]_i \times [\text{UA}]_i \times B_{\{0,i\}} \\ &= i^2 \times i \times i^3 = i^6 \text{ EXACT} \end{aligned}$$

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This is the closed structural derivation of the i^6 power law — **not assumed, but derived from how three layered vacuum quantities scale with the integer layer index.**

UQFF Closed Identity

For each layer $i \in \{1, 2, \dots, 26\}$:

$[\text{SCm}]_i = i^2$ (SCm density at layer i — quadratic scaling)

$[\text{UA}]_i = i$ (UA gradient at layer i — linear scaling)

$B_{\{0,i\}} = i^3$ (background magnetic field at layer i — cubic scaling)

$$w_i = [\text{SCm}]_i \times [\text{UA}]_i \times B_{\{0,i\}}$$

$$= i^2 \times i \times i^3$$

$$= i^{(2+1+3)}$$

$$= i^6 \text{ EXACT}$$

$$A_{26} = \sum_{i=1}^{26} w_i = \sum_{i=1}^{26} i^6 = 1,307,797,101 \text{ EXACT}$$

Physical Interpretation

The three exponents (2, 1, 3) are themselves structurally meaningful:

- **2 = D_phys** – 2 (transverse SCm density scaling)

- **1 = linear UA-channel gradient**

- $3 = D_{\text{phys}} - 1 = \text{SU}(3) \text{ channel count}$ (magnetic background)

The sum $2 + 1 + 3 = 6 = D_{\text{BSFG}}$, the bulk-edge dimension. Layer weight w_i thus scales as $i^{D_{\text{BSFG}}}$, providing a deep structural link between the integer primitives D_{phys} , D_{BSFG} , and the amplification mechanism.

Predictive Consequence

Predicts that any UQFF amplification cascade summing over D_{crit} layers with field exponents totaling D_{BSFG} will produce a $\Sigma i^{D_{\text{BSFG}}}$ sum. Other amplification factors using different exponent combinations should be searchable in the UQFF whitepaper corpus.

NOT REPLACEMENT

SM has no analogous decomposition of vacuum fields into integer-power layer weights. UQFF supplies a complete structural account.

Reference

- Source: PAPER_1155 DPM 26-Layer Amplification Particle Masses
- Related: PAPER_1510 ($A_{26} = 1,307,797,101$), PAPER_062 (DPM lattice)
- Calculator dispatch: `calculate_paradox({"paradox": "dpm_layer_weight_i6_decomp"})`

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